LAB – 0

Setting up your development environment

Lab Overview:

For this lab, we will be setting up our Linux development environment for systems programming in C. Make sure to follow all steps outlined in the “tasks” subsection of this lab description. This setup should be done as early as possible. There may be many different issues depending on your machine, and it is better to resolve them earlier rather than later. Be sure to ask questions if anything is not immediately clear.

Core Tasks:

1. Set up your virtual machine (VM) and install Ubuntu 18.04.5 LTS (Bionic Beaver)
2. Install development tools (GDB, Valgrind, IDE)
3. Hello World in C. (developing, compiling, running, and testing code)

Task Details:

Be sure to follow these instructions while setting up your VM. We will be grading your work on a VM running this exact setup, so if these instructions are not followed, then there is a significant chance that your submissions will not compile or run when we grade them. This is the most important lab because the rest of your lab and project submission depends on it.

Important note:

A BIOS setting is required for Windows users. Check out this link to change your BIOS settings if your VM does not run. https://www.youtube.com/watch?v=m7cHUNXB7j4

For Mac users, go to Security&Privacy and check "detailed" shows below "Allow apps downloaded from" and restart your computer.

UTM: Another option for Mac users is UTM, which can be downloaded for free from the official UTM website. Once you have downloaded the AMD image and UTM, you can use UTM to then emulate the image, and then you will be prompted in a shell with a couple of options, but pick

install ubuntu and you should be good to go. NOTE: if you get an error “Display Output is not active” follow these instructions https://github.com/utmapp/UTM/discussions/5555. Also, this is runs extremely slow.

If you own the newest Apple M1, you need to install “Parallels Desktop 16”. DO NOT FOLLOW the installation part in this document. Follow this link; however, install 18.04.5 instead of 20.04. https://youtu.be/FzkLILpNGZw

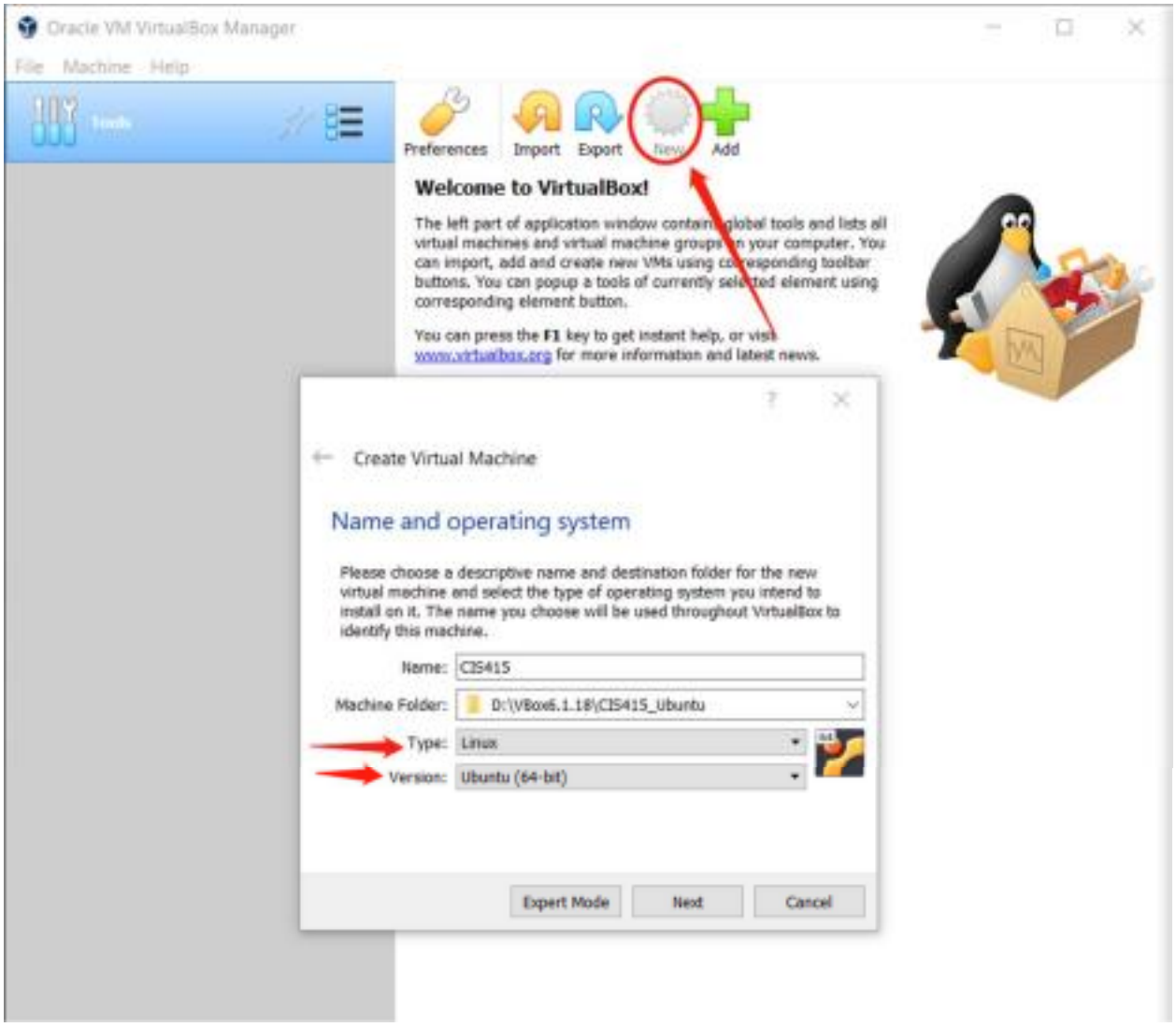
Extra help for setup(MAC) - https://youtu.be/1PL-0-5BNXs?si=xSHNjasQIL_dR6P8

Downloading files (VM).

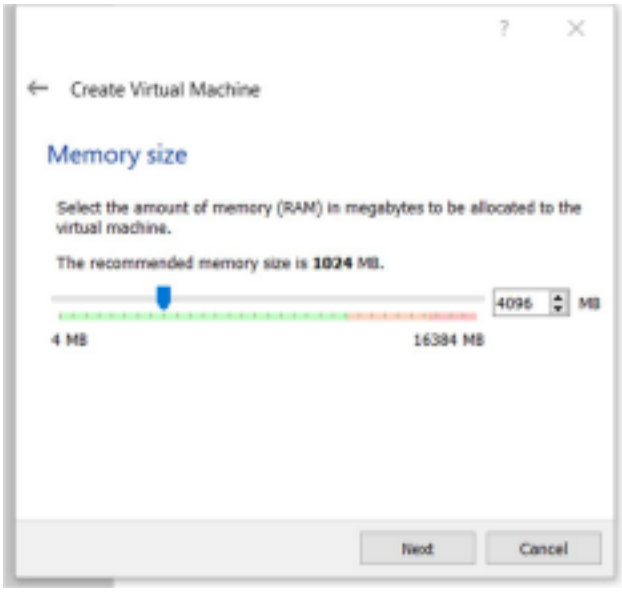
1. Download and install Oracle Virtual Box.
    a. Link (https://www.virtualbox.org/)
2. Download the “64-bit PC (AMD64) desktop image” Linux ubuntu18.04.5 (Bionic Beaver) ISO
    a. Link (https://releases.ubuntu.com/18.04.5/?_ga=2.79050774.325258749.1616538386-954279093.1616538386)
    b. Make sure you download the Desktop version

Setting up your VM.
(Most of the instructions are for Windows 10; there should be only some minor differences compared to Mac)
Once you have the ISO downloaded and VirtualBox installed, open VirtualBox.

- Click on “New”, then enter the name “CIS-415”.
- Change the machine folder to a location of your choice (make sure you have at least 15GB of space).
- The Type is Linux.
- The version is Ubuntu (64bits) (If you have a 32-bit machine, which is very unlikely, please contact us)
- Click on Next



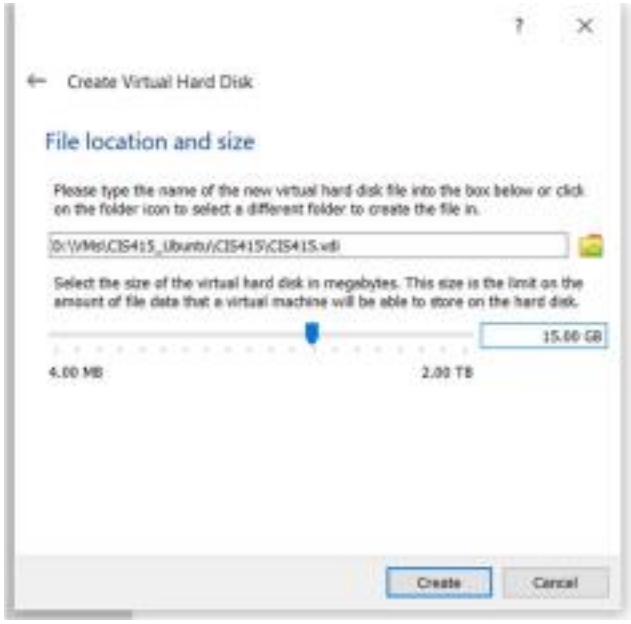
● Select the amount of RAM to use (we used 4GB of RAM. It's best to use a good amount so the VM runs smoothly, so if you have more to spare, you may add more. Click on Next.



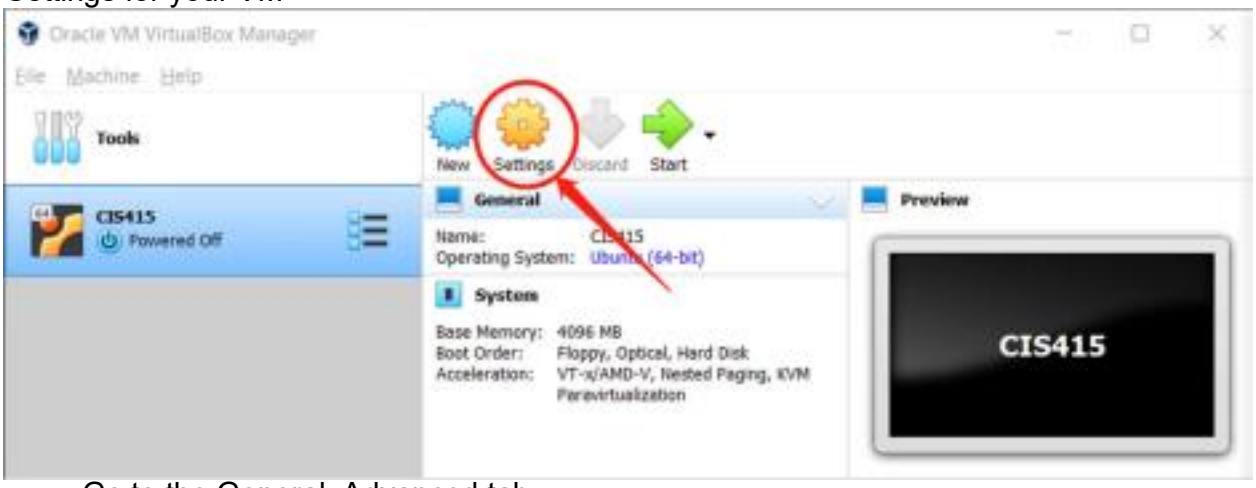
● Check the “create virtual disk now” (which should be the default), then click “create”.

● You should get three pop-up windows

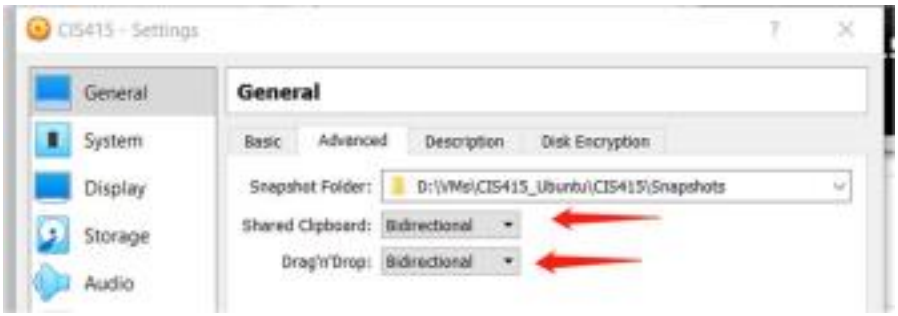
- ○ Select “VirtualBox Disk Image.”
- ○ Select “Dynamically allocated”.
- ○ Assign 15 GB of memory. (Ubuntu requires at least 10G+ of storage space to install, so make sure you give the VM enough memory)
- ○ Click Create.



Settings for your VM

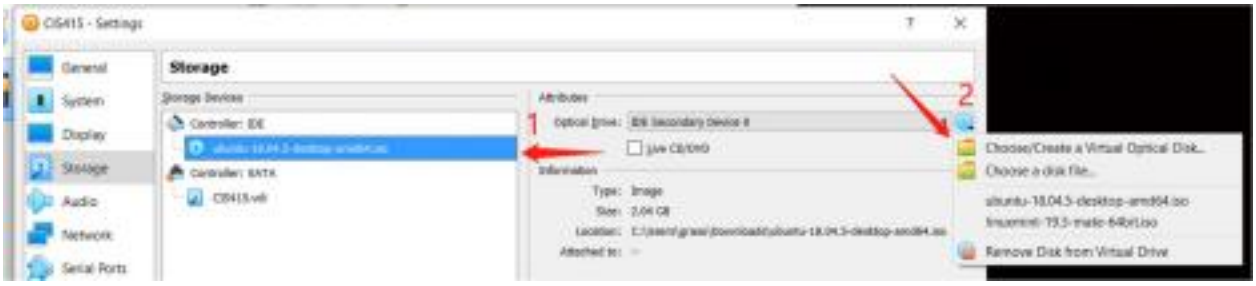


- Go to the General, Advanced tab
  - Change both “Shared Clipboard” and “Drag’ and ’Drop” to bidirectional



- Go to System, Processor tab
  - Change processor to at least 2 (this will be important for the last project)

- Next go to Storage
  - Click on the disk at location 1
  - Then edit the iso image you want to choose at location 2
  - Select the ubuntu 18.04.5 image you downloaded earlier



Now, start your VM and Install Ubuntu

- Select your VM on the left-hand side and click “Start.”
    - First time boot up will take a little while

- Select English and Install Ubuntu

- For the next few windows, select the following options
    - Keyboard layout
        - English (US), English (US)) You can add other keyboard installations afterwards if you want to.
    - Updates and other software
        - Select “Normal installation.”
        - Check “Download updates while installing Ubuntu.”
        - Check “install third-party software …”

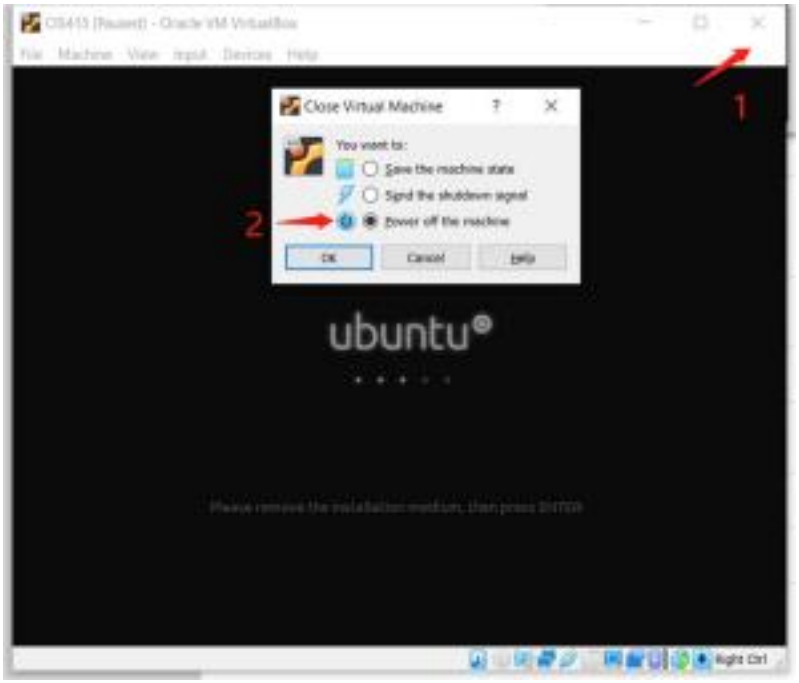
- ○ Installation type
  - ■ Check and only check “Erase disk and install Ubuntu.”
  - ■ Click on install now
  - ■ For the pop-up window (Write the changes to disks?), click “Continue”. ○ Choose a location
- ○ Put in your name, username, and a password, and click “Continue”.

● The installation will take a little time, so just be patient. Wait and relax.

● After installation, click “Restart Now”.

● When you see this window, “Please remove the installation medium, then press ENTER.”

- ○ Close the window at location “1.”
- ○ Select “power off the machine” at location “2.”
- ○ Click “OK.”



● Boot up your VM

- ○ If a window pops up and reminds you to upgrade to 20.04, decline.
- ○ ***Don’t upgrade! ***

● Let's get the system up to date

- ○ Press “control+shift+t” or right click on the desktop and click “Open Terminal”
- ○ Type in the following commands (using sudo commands will prompt a password request)
  - ■ $ sudo apt update
  - ■ $ sudo apt upgrade
- ○ Wait for the update to finish (you can change your window size by going into ubuntu settings, search “Display”, and then change the resolution)

Extra tip on setting up shared folder
https://askubuntu.com/questions/456400/why-cant-i-access-a-shared-folder-from-within-my virtualbox-machine
You can follow this link to install the VBox guest version and set up a shared folder; this is not required.

Setting up our development environment

- Bring up the terminal and install development tools with the following commands
  - $ sudo apt install build-essential
  - $ sudo apt install valgrind
- To check if you have successfully installed the developing tools, use the following command
  - $ gcc --version
- If the terminal prompt "command 'gcc' not found," then you have to do the above step again.

IDE
You may use any IDE you like. Examples are VI, Eclipse, VSCode, etc.

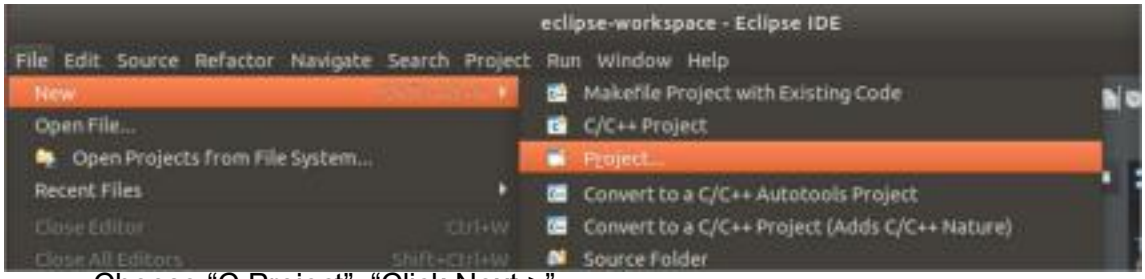
Some tips on Eclipse (optional)

Eclipse contains built-in GDB GUI and Valgrind which is convenient for developing C projects.

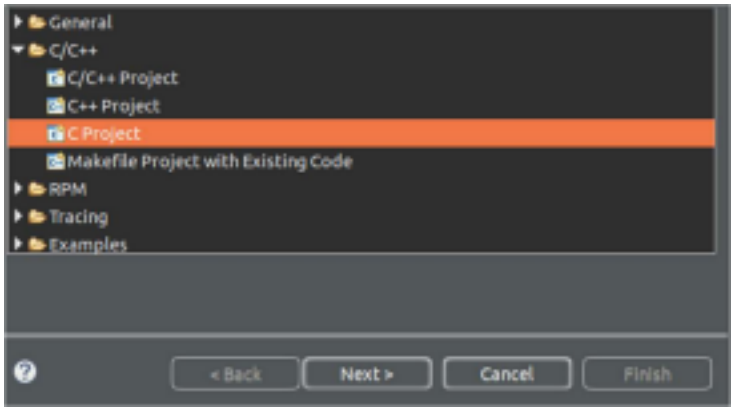
- Installing Eclipse
  - Go to link and download "Eclipse IDE 2021-03" (https://www.eclipse.org/downloads/packages/release/helios/m7/eclipse-ide-linux developers)
  - Extract the tar file, open the directory and double click "eclipse-inst"
  - Select "Eclipse IDE for C/C++ Developers"
    - Leave everything as default and install

Now let's write a basic C program

- Start your Eclipse
  - Click on "File"
  - Choose "Project…"



- Choose “C Project”, “Click Next >”.

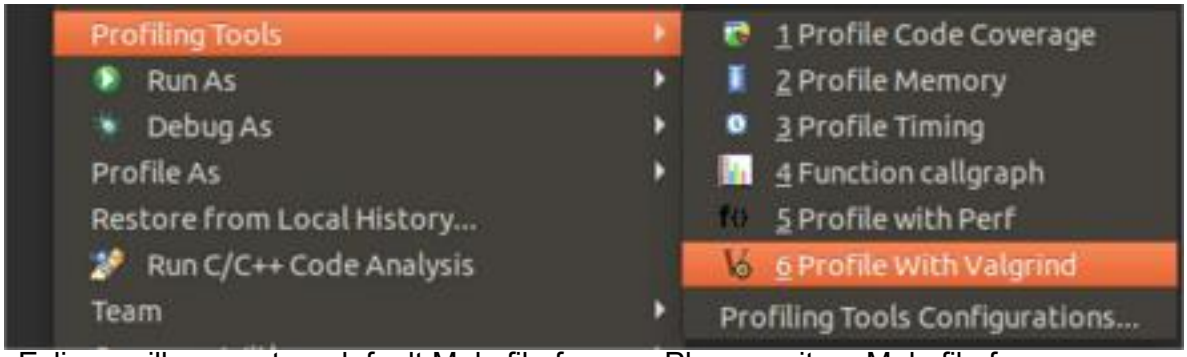


- Put “Lab0” in project name
- Choose “Hello World ANSI C Project” and “Linux GCC” on right

- Then click “Finish”
- Click on the hammer icon on top to build
- Click on the green play icon to run

● You can also run “valgrind” to check memory leak in Eclipse

- Right click on project “Lab0”
- Select “profiling Tools”
- Profile with Valgrind



● Eclipse will generate a default Makefile for you. Please write a Makefile for your project. ○ For future projects and labs, you will also be writing your own Makefiles ● ***Your target executable should be named "Lab0.out"***

SUBMISSION:

- What to Submit
  - A tar or zip file including:
    - Lab0.c (this file only prints hello world)
    - Makefile (Your target executable should be named "Lab0.out")
- The naming convention of the zip/tar file while uploading to Canvas

  UoID_duckID_LAB/ProjectX
    - UOID: alex
    - DuckID: 951505xxx
    - Submission for: Lab0
    - So, the name of the zip/tar file should be:
      - alex_951505xxx_Lab0.tar or alex_951505xxx_Lab0.zip
- How We are Going to Test Your Program
  - Open a terminal where your src code is
  - Type command (your src code and Makefile must be in the same directory)
    - $ make
    - $ ./Lab0.out
  - To test for memory leak type
    - $ valgrind ./Lab0.out